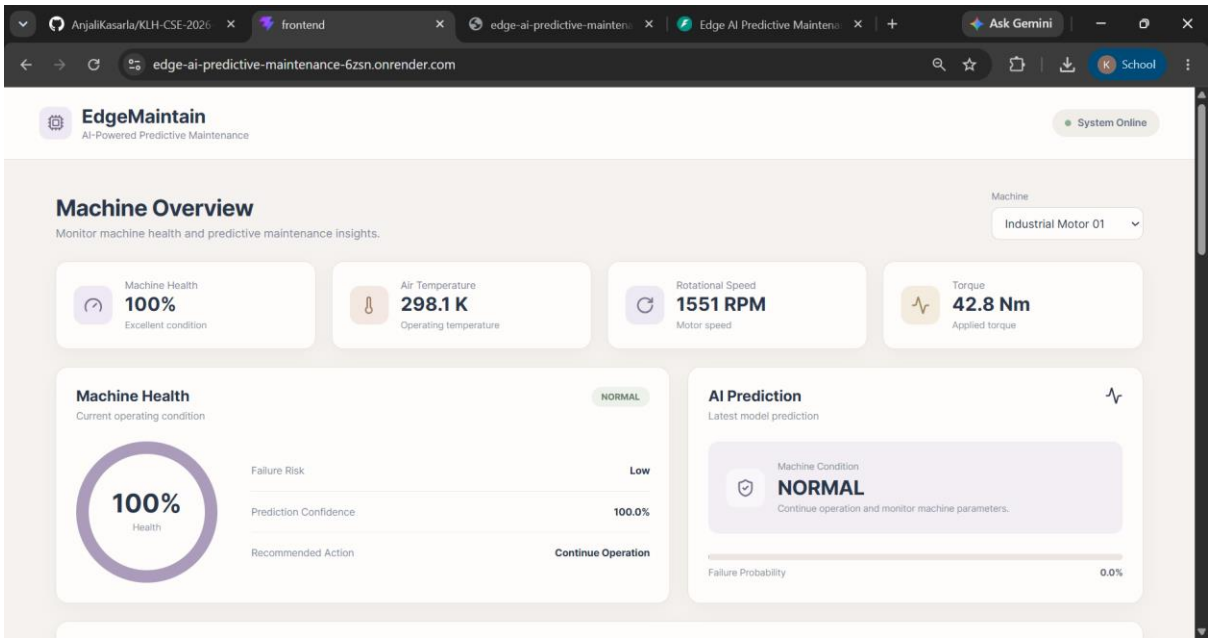
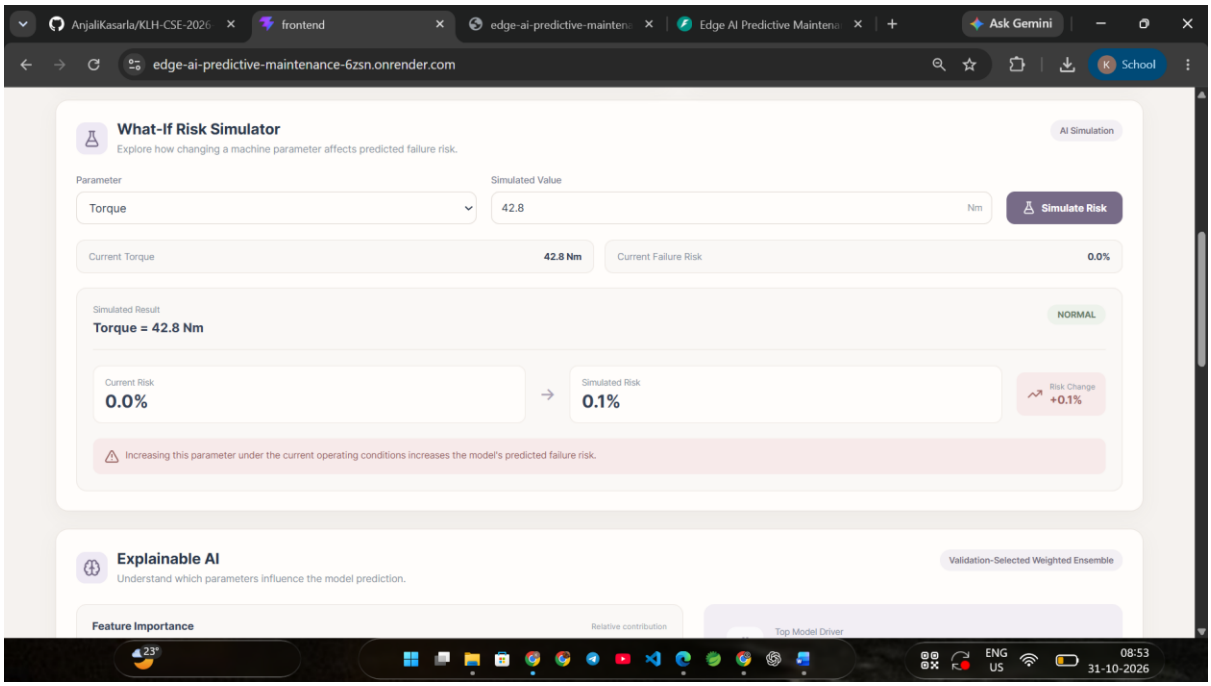
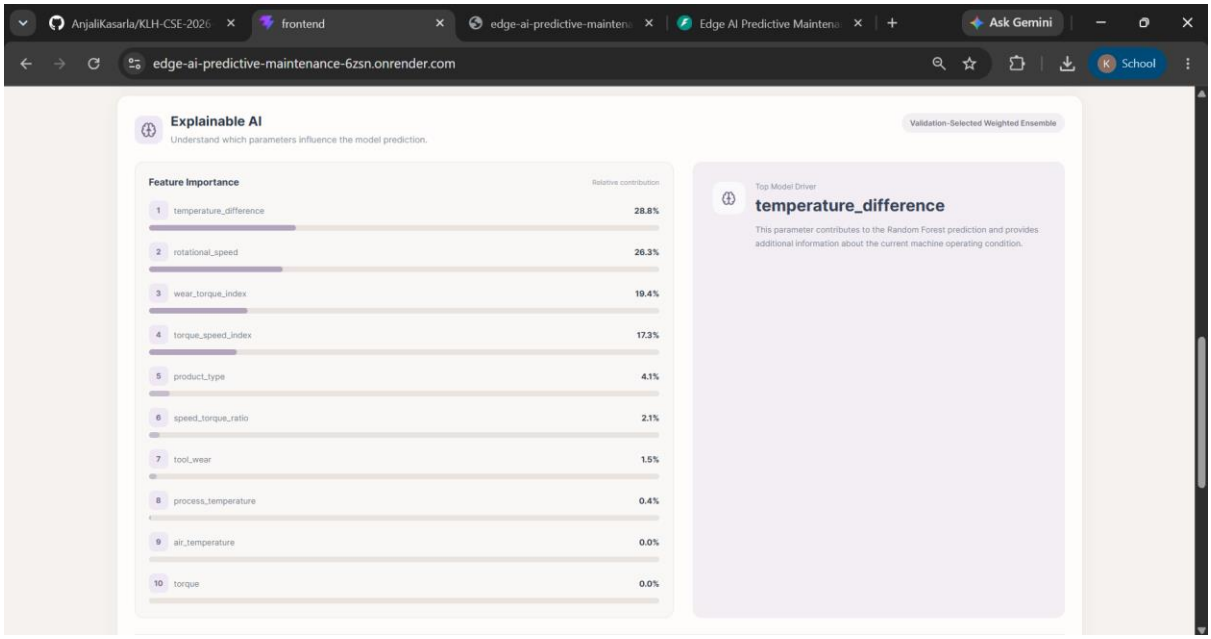
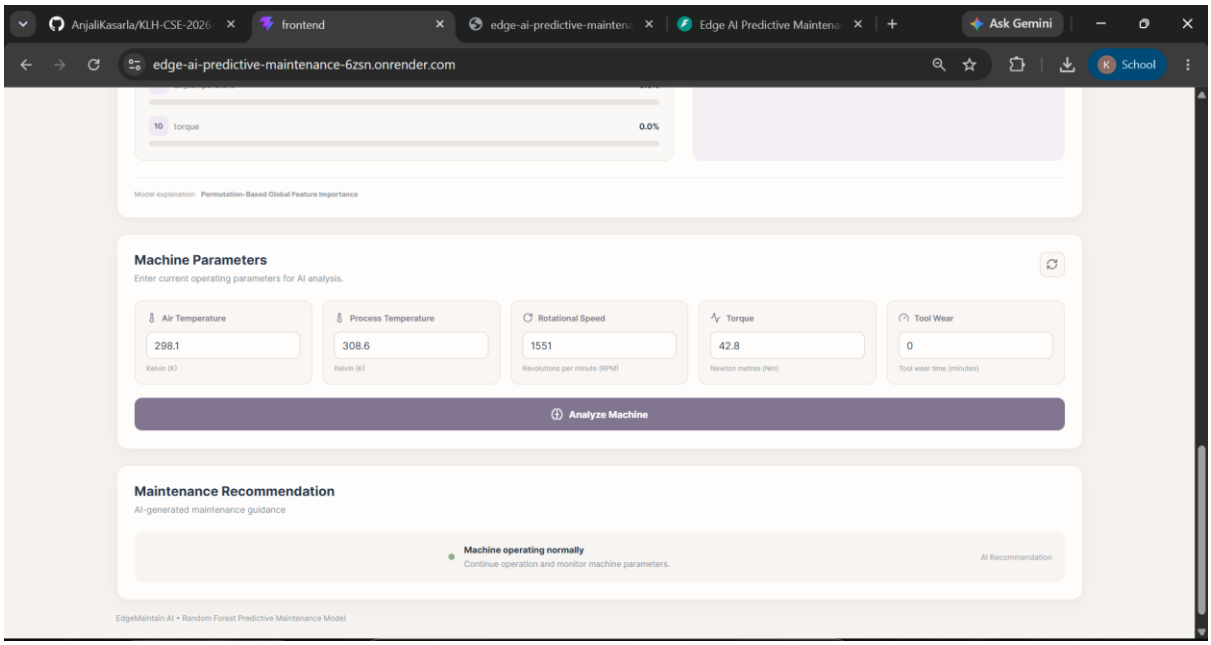
Frontend Application:

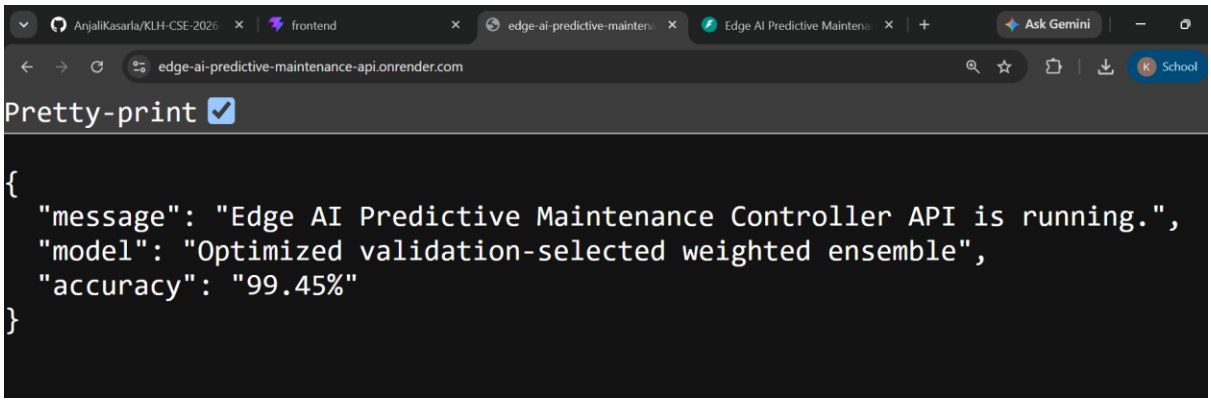





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frontend
edge-ai-predictive-maintena
Edge AI Predictive Maintena
Ask Gemini
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School
Explainable AI
Understand which parameters influence the model prediction.
Validation-Selected Weighted Ensemble
Feature Importance
Relative contribution
1 temperature_difference 28.8%
2 rotational_speed 26.3%
3 wear_torque_index 19.4%
4 torque_speed_index 17.3%
5 product_type 4.1%
6 speed_torque_ratio 2.1%
7 tool_wear 1.5%
8 process_temperature 0.4%
9 air_temperature 0.0%
10 torque 0.0%
Top Model Driver
temperature_difference
This parameter contributes to the Random Forest prediction and provides additional information about the current machine operating condition.


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School
10 torque 0.0%
Model explanation: Permutation-Based Global Feature Importance
Machine Parameters
Enter current operating parameters for AI analysis.
Air Temperature
298.1
Kelvin (K)
Process Temperature
308.6
Kelvin (K)
Rotational Speed
1551
Revolutions per minute (RPM)
Torque
42.8
Newton metres (Nm)
Tool Wear
0
Tool wear time (minutes)
Analyze Machine
Maintenance Recommendation
AI-generated maintenance guidance
Machine operating normally
Continue operation and monitor machine parameters.
AI Recommendation
EdgeMaintain AI • Random Forest Predictive Maintenance Model

Backend API:



API Documentation:

